MIL-C-6100A(AS)
15 October 1973
SUPERSEDING
MIL-R-6100(Aer)
15 April 1950

MILITARY SPECIFICATION

CONNECTOR, RECEPTACLE, ELECTRICAL, AIRCRAFT

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE AND CLASSIFICATION

1.1 Scope. The receptacles covered by this specification shall be designed for aircraft applications. The receptacle shall be a two-wire connector designed to mate with plug Bureau of Ordnance Drawing No. 454642, or equivalent, by a spring twist-lock action, and shall be furnished with two MIL-W-5086/1 cables, attached and identified as "GND" and "+" respectively. Both leads shall be terminated in MS25181-2 permanent splices, or equivalent.

1.2 Classification. The receptacles covered by this specification shall be in accordance with MS 25250.

1.3 Precedence. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

a. Contract. The contract shall have precedence over any specification or drawing.

b. This specification. This specification shall have precedence over all applicable subsidiary specifications, publications, and drawings of section 2 herein. Any deviation from this specification, or from the referenced specifications, publications, or drawings where applicable, must be specifically approved in writing by the Naval Air Systems Command.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

FSC 5935

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SPECIFICATIONS

Federal

QQ-A-225/5	Aluminum Alloy, Bar, Rod and Wire, Rolled, Drawn or Cold Finished, 2017
QQ-A-591	Aluminum Alloy Die Castings
QQ-B-637	Brass, Naval, Rod, Wire, Shapes, Forgings, and Flat Products with Finished Edges (Bar, Flat Wire, and Strip)
QQ-C-530	Copper-beryllium Alloy Bar, Rod, and Wire (Copper Alloy Numbers 170 and 172)
QQ-W-470	Wire, Steel, Carbon, Spring, Music
VV-P-236	Petrolatum, Technical

Military

MIL-M-14	Molding Plastics and Molded Plastic Parts, Thermosetting
MIL-W-5086/1	Wire, Electric, Hookup and Interconnecting, Polyvinyl Chloride Insulation, Nylon Jacket, Tin-coated Copper Conductor, 600 Volt, 105 Deg. C
MIL-S-6715	Springs, Helical, Aircraft
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-P-19468	Plastic Rod, Polytetrafluoroethylene, Molded and Extruded
MIL-P-19833	Plastic Molding Material and Plastic Molded Parts, Glass Fiber Filled, Diallyl Phthalate Resin
MIL-C-55330	Connector, Preparation for Delivery of

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STANDARDS

Federal

Fed. Test Method Std. No. 601	Rubber, Sampling and Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-454	Standard General Requirements for Electronic Equipment
MS25181	Splice, Electric, Permanent, Crimp Style Copper, Insulated, Type II, Class 1
MS25250	Receptacle, Electric Connector, Rocket Igniter Assembly

DRAWINGS

Bureau of Ordnance

454642	Electrical Connector Jack Plug

(Available through the Naval Air Systems Command)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

B 117-64	Standard Method of Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Qualification. The connectors furnished under this specification and MS 25250 shall be products which are qualified for listing on the applicable Qualified Products List (QPL) at the time set for the opening of bids (see 4.3 and 6.4).

3.2 Materials. The materials used in the manufacture of receptacle connectors shall be as specified herein; however, when a definite material is not specified, a material shall be used which will enable the connectors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty for acceptance of the finished product.

3.2.1 Plastic materials. Unless otherwise specified in the applicable Military Standard, the plastic materials specified in MIL-M-14, MIL-P-19468, and MIL-P-19833 shall be used.

3.2.2 Shells and springs. Shells and springs or any other part of the receptacle that uses metals shall be in accordance with QQ-A-225/5, QQ-A-591, QQ-B-637, QQ-C-530, QQ-W-470, and MIL-S-6715.

3.2.3 Rubber goods. Unless otherwise specified, rubber used in the manufacture of receptacles shall meet the test requirements of Fed. Test Method Std. No. 601.

3.2.4 Threaded parts. Threaded parts of the receptacle shall conform to the requirements of MIL-S-7742.

3.2.5 Dissimilar metals. When dissimilar metals are employed in intimate contact with each other, suitable protection against electrolytic corrosion shall be provided as specified in Requirement 16 of MIL-STD-454.

3.2.6 Corrosion resistance. Materials shall be of a corrosion-resisting type or suitably processed to resist corrosion.

3.2.7 Soldering. Only noncorrosive fluxes shall be used, unless it can be shown that corrosive elements have been satisfactorily removed after soldering. All excess flux shall be removed.

3.2.8 Substitution of material. If the receptacle contractor desires to substitute for a specified or previously approved material or fabricated part, he shall submit to the procuring activity, with his bid or sample, a statement to that effect, describing the claim that such substitute is suitable. Similar notification and substantiating evidence shall be submitted at any later time if such change again becomes

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necessary or desirable. At the discretion of the procuring activity, test samples may be required to prove the suitability of the proposed substitute.

3.2.9 Finish. Shells and nuts may be finished by light sand blasting and clear lacquer or other process which produces a smooth clean surface.

3.3 General design requirements.

3.3.1 General. The rocket igniter receptacle shall be designed to meet the requirements specified herein.

3.3.2 Design and construction. The receptacle shall be of the design and construction and of the physical dimensions specified in MS 25250.

3.3.3 Seizing of the nut. To prevent seizing of the nut on the receptacle, a thin coating of dielectric compound shall be applied to the threads of the receptacle. The compound shall be petrolatum conforming to VV-P-236.

3.3.4 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance.

3.3.5 Identification marking. The receptacles shall be marked with the part number and the manufacturer's name or trademark.

3.3.6 Weight. The receptacle shall not weigh more than 0.09 pound.

3.3.7 Performance. The receptacle shall satisfy the performance requirements as specified in the paragraphs headed as follows:

Examination of Product (4.7.1)
Continuity (4.7.2)
Millivolt Drop (4.7.3)
Torque (4.7.4)
Insulation Resistance (4.7.5)
Thermal Cycling (4.7.6)
Vibration (4.7.7)
Shock (4.7.8)
Durability (4.7.9)
High Potential (4.7.10)
Humidity (4.7.11)
Corrosion Resistance (4.7.12)

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3.3.8 Workmanship. Receptacles shall be manufactured and processed in a careful and workmanlike manner, in accordance with good design and practice. Machined surfaces shall be smooth and free from rough edges and burrs.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

a. Qualification inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

c. Inspection of preparation for delivery (see 4.5)

4.3 Qualification inspection. Qualification inspection shall consist of all the tests specified in 4.7 and shall be conducted in the order in which they are listed.

4.3.1 Qualification samples. Qualification inspection samples shall consist of ten connectors. Two reproducible copies of outline and assembly drawings shall be submitted with the samples. The samples shall be plainly identified by a securely attached, durable tag marked with the following information:

Sample for Qualification Test
Receptacle, Rocket Igniter
MS 25250
(Manufacturer's Part Number)
(Name of Manufacturer)
Submitted by (name) (date) for qualification test
in accordance with requirements of Specification
No. MIL-R-6100A(AS) under (reference authorizing
letter)

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4.4 Quality conformance inspection. Quality conformance inspection shall consist of groups A, B and C inspections.

4.4.1 Group A inspection. Group A inspection shall consist of the following test methods:

a. Examination of product (see 4.7.1)

b. Continuity (see 4.7.2)

Group A sampling shall be in accordance with MIL-STD-105, general inspection Level II. The Acceptance Quality Level (AQL) shall be .65 for major defects and 2.5 minor defects.

4.4.2 Group B inspection. Group B inspection shall consist of the following test methods:

a. Torque (see 4.7.4)

b. Insulation resistance (see 4.7.5)

c. High potential (see 4.7.10)

Group B samples shall have passed Group A inspection. Group B sampling shall be in accordance with MIL-STD-105, special inspection level S-4. The AQL shall be 2.5.

4.4.3 Group C inspection. Group C samples shall be selected from inspection lots which have passed Group A and Group B inspections and shall be subjected to the following test methods:

a. Subgroup 1
   Millivolt drop (see 4.7.3)
   Durability (see 4.7.9)

b. Subgroup 2
   Vibration (see 4.7.7)
   Shock (see 4.7.8)

4.4.3.1 Group C sampling plan. Four sample connectors, representative of the production units, shall be selected every three months after notification of qualification. The units shall be equally divided between the two subgroups. Upon passing this inspection two successive times, the supplier may select units every six months. In event of a failure, sampling shall revert to three month intervals.

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4.4.4 Non-compliance. If a sample fails to pass Group C inspection, the supplier shall take corrective action on the material or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of related lots shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, Group C inspection shall be repeated on corrected samples only for the groups in which failures were encountered. Groups A and B inspections may be re-instituted; however, final acceptance shall be withheld until the Group C re-inspection has shown that the corrective action was successful.

4.5 Inspection of preparation for delivery. Samples from a quantity of shipping containers fully prepared for delivery, just prior to closure, shall be selected at random from each lot in accordance with MIL-STD-105 at Inspection Level S-2. The lot size shall be the number of shipping containers in the lot. Each sample selected shall be examined for defects and to determine conformance to all the applicable requirements of Section 5 of this specification. If the number of defective shipping containers exceeds the AQL of 4.0, the lot represented by the sample shall be rejected.

4.6 Test conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of approximately 29.92 inches of mercury and a temperature of approximately 25° C (77° F). Where tests are made with atmospheric pressure or temperature substantially different from the above values, proper allowance for change in instrument reading shall be made for the difference from the specified conditions.

4.7 Test methods.

4.7.1 Examination of product. Each receptacle submitted for tests shall be carefully examined to determine conformance with this specification with respect to material, workmanship, design and construction, marking, and conformance to detail drawings.

4.7.2 Continuity. Both leads of each receptacle shall be checked for continuity with a suitable test device wired in the circuit.

4.7.3 Millivolt drop. The millivolt drop shall be secured at an ambient temperature of 25 ±5° C (77 ±9° F) between receptacle and corresponding plug when assembled as in service. The millivolt drop shall be measured with each contact carrying a current of 11 amperes and after the contacts have been carrying this current for one hour. Precision laboratory instruments and techniques shall be used in measuring the millivolt drop. The millivolt drop shall not exceed 20 millivolts for the positive contact and 25 millivolts for the ground contact when measurements are secured at the tips of the solder cups of the receptacle and plug as shown

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in Figure 1 (A and B). The millivolt drop shall not be greater than 100 millivolts for the positive lead and 85 millivolts for the ground lead when measurements are secured across the entire connector, from the splice crimp of the receptacle to the end of the cable on the plug side as shown in Figure 1 (C and D).

4.7.4 Torque. To unlock the plug from the receptacle, a maximum of 8 pound-inches and a minimum of 2 pound-inches twisting force shall be applied to the plug in a counterclockwise direction tending to loosen the plug from the receptacle.

4.7.5 Insulation resistance. The initial insulation resistance of the receptacle between the ground and the positive side of the circuit shall not be less than 100,000 megohms when the measurement is made between the tips of the splices with a megohm bridge. The reading shall be secured in an ambient temperature of 25 ±3° C (77 ±5° F) after the sample has been conditioned to this temperature for at least one hour.

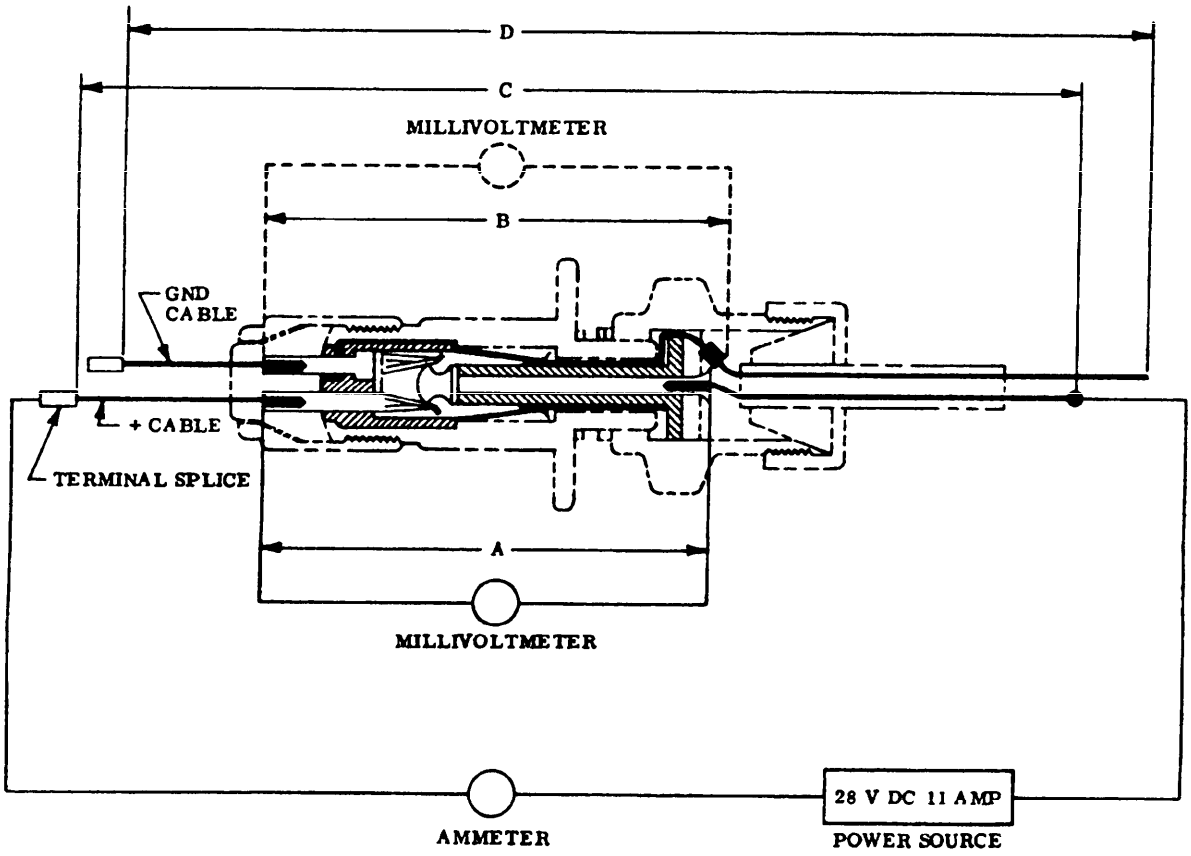


Figure 1. Wiring Diagram for Millivolt Drop Test

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4.7.6 Thermal cycling. The receptacle shall be exposed to at least five cycles of temperature change as indicated in Table I. The receptacle shall be held at the specified maximum and minimum temperatures for not less than 30 minutes. The rate of temperature change within the temperature chamber shall be not less than 2° C (3.6° F) each minute. The receptacle may be transferred from one chamber to another for these changes, in which case the sample shall be maintained at room temperature for a maximum of 15 minutes or a minimum of 5 minutes between exposures to the extreme temperature conditions.

TABLE I

Operation	Temperature
Start at	+85° C (+185° F)
Reduce to	-55° C (-67° F)
Return to	+20° C (+68° F)

4.7.7 Vibration. The receptacle sample shall have 12-inch lengths of cable attached to its terminals. The sample shall be mounted in a vertical position on the vibration table with the receptacle opening facing downward and the ends of the cables secured to a stationary support, with a suitable test device wired in the circuit to determine that the vibration does not cause the circuit to be momentarily open. A mating plug, BuOrd Drawing No. 454642, or equivalent, shall be mated with the receptacle with its cables shorted to complete the circuit. The sample shall be vibrated for a period of 2 hours in a horizontal plane and 2 hours in a vertical plane. The vibration shall consist of a total amplitude of 0.060 inch at a cycling frequency of 10 to 55 to 10 cycles per second, the cycle frequency to be accomplished in 1 minute. There shall be no mechanical or electrical failure as determined by visual examination, and the electrical continuity of the circuit shall not be interrupted during the vibration test.

4.7.8 Shock. The sample shall be mounted on the shock device or carriage. The sample shall be subjected to approximately 1/2 sine wave transient shock impulses of 50 gravity units and a duration of 11 ±1 milliseconds. One shock shall be applied in each direction of the three major axes of the connector. The receptacle shall be mated with a plug, BuOrd Drawing No. 454642, or equivalent and held by normal locking means only. All contacts shall be wired in series, and a current of 100 ±10 milliamperes shall flow through the series circuit during the test. The attached wires shall be clamped to structures that move with the connectors. A minimum of 8 inches of wire or cable shall be unsupported behind the rear of each connector. A suitable instrument shall be employed to indicate any discontinuity exceeding 10 microseconds in the circuit.

4.7.9 Durability. Five receptacles shall be subjected to 500 cycles of insertion and withdrawal with a mating plug, BuOrd Drawing No. 454642 or equivalent.

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The rate of insertion and withdrawals shall not exceed 500 cycles per hour. After durability testing, the average millivolt drop, as compared to the initial millivolt drop, shall not have increased more than 5 millivolts for the positive lead and 10 millivolts for the ground lead when measurements are secured across the entire connector, or at the tips of the solder cups as shown in Figure 1 (A, B, C, and D).

4.7.10 High potential. The receptacle shall show no breakdown or flashover when a voltage of 2,000 volts peak (1,414 volts rms), 60 cycles, is applied between the positive lead and ground for a period of 1 minute. The test voltage shall be applied gradually at a rate of approximately 500 volts each second until the specified voltage is reached.

4.7.11 Humidity. The receptacle with mated plug shall be exposed to a temperature of 71 ±2° C (160 ±3.6° F) and relative humidity of 95 ±5 percent for a period of 7 days. Immediately after exposure to the humidity test, and without forced drying, the assembled connector shall withstand 117 volts rms, 60 cycles, between the positive and ground cables for a period of 5 minutes. There shall be no mechanical or electrical failure.

4.7.12 Corrosion resistance. The receptacle and mated plug, BuOrd Drawing No. 454642, or equivalent, shall be subjected to the test conditions outlined and in accordance with ASTM B 117-64, for 100 hours. Immediately after this test, the receptacle and mated plug shall be washed with tap water for 5 minutes, after which the assembly shall be maintained at an ambient temperature of 57.2 ±1° C (135 ±1.8° F) for a period of 3 hours. The millivolt drop measurements shall then be secured between the splice crimp of the receptacle and the plug cable end as shown in Figure 1 (C and D). The millivolt drop shall not be greater than 200 millivolts for the ground lead and 150 millivolts for the positive lead. After this test, the receptacle shall be examined for corrosion of its parts, and the plug shall not be corroded to the receptacle or be difficult to disengage. Following the corrosion test, the plug shall be submitted to 10 cycles of insertion and withdrawal, after which the millivolt drop measurement shall be secured. The values obtained shall not exceed 150 millivolts for the ground lead and 100 millivolts for the positive lead.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging, packing and container marking. Preservation and packaging, packing and container marking shall be in accordance with MIL-C-55330.

6. NOTES

6.1 Intended use. The connector covered by this specification is intended for use in aircraft in applications such as electrical connectors to the individual rockets from igniter circuits, and applications with bomb racks, bomb harness assemblies and missile adapter packages.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Title, number and date of the applicable MS standard and the part number.

c. Levels of preservation and packaging and packing in accordance with MIL-C-55330 (see 5.1).

6.3 Voltages. Voltages used in high potential and humidity tests shall be considered to be peak values.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List (QPL), whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification.

6.4.1 Application for qualification. The activity responsible for the QPL for MIL-C-6100 is the Naval Air Systems Command, The Naval Ammunition Depot, Crane, Indiana, has been designated by the Naval Air Systems Command as agent for the establishment of the QPL. Requests for information pertaining to and applications for qualification should be addressed to:

Commanding Officer
Naval Ammunition Depot
Crane, Indiana 47522
Attention: Code 70515

Preparing Activity:
Navy - AS
(Project No. 5935-N887)

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE MIL-C-6100A(AS)
CONNECTOR, RECEPTACLE, ELECTRICAL, AIRCRAFT

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

TELEPHONE NO.

DATE

FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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